FOUNDRY TECHNOLOGY NOTES

Pouring, Gating, Risers, Chills and Special Casting Processes

Reference framework: Hajra Choudhury - Elements of Workshop Technology, Vol. I (Manufacturing Processes)

1. Optimum Pouring Temperature

Pouring temperature is the temperature of molten metal at the instant it is poured into the mould. It is normally kept above the liquidus temperature so that the metal remains sufficiently fluid while travelling through the gating system and filling the mould cavity.

1.1 Meaning of optimum pouring temperature

The optimum pouring temperature is not simply the highest possible temperature. It is the temperature that gives adequate fluidity and complete mould filling while avoiding excessive oxidation, gas absorption, mould attack, coarse grains, shrinkage difficulties and unnecessary energy consumption.

- Too low a pouring temperature may cause misrun, cold shut, incomplete filling and poor reproduction of thin sections.
- Too high a pouring temperature may increase oxidation, gas pickup, metal-mould reaction, penetration, shrinkage tendency and solidification time.
- The correct value depends on alloy composition, casting size and section thickness, mould material, distance travelled by the metal, gating design and required surface finish.

1.2 Superheat

The difference between the pouring temperature and the liquidus temperature is called superheat.

Superheat = Pouring temperature - Liquidus temperature

Factor	Effect on selected pouring temperature
Thin and intricate casting	Usually needs higher fluidity and therefore comparatively higher superheat.
Large, heavy section	May need less superheat than a very thin casting because the section loses heat more slowly.
Long flow path / complicated gating	More heat is lost before cavity filling; adequate superheat is required.
Sand mould	Permits a broad range but very high temperature may cause burn-on or penetration.
Metal mould	Rapid heat extraction requires careful temperature control.
Alloy with poor fluidity	Generally requires more superheat than a highly fluid alloy.

EXAM POINT: Optimum pouring temperature is a compromise: it must be high enough for complete filling, but not so high that casting defects and metallurgical deterioration increase.

2. Pouring Time and Pouring Rate

2.1 Pouring time

Pouring time is the total time required to fill the mould cavity with molten metal. It is an important design quantity because the mould must fill before premature solidification blocks the flow, but the filling should not be so violent that turbulence and erosion occur.

2.2 Pouring rate

Pouring rate is the quantity of molten metal entering the mould per unit time.

Average pouring rate = Weight (or volume) of metal poured / Pouring time

- A very slow rate can lead to misrun, cold shut and excessive temperature loss.
- A very fast rate can cause turbulence, aspiration of air, mould erosion, sand inclusions and excessive impact on the cavity.
- The required rate depends on casting mass, alloy fluidity, minimum section thickness, mould material, gating design and permissible turbulence.

2.3 Factors governing pouring time

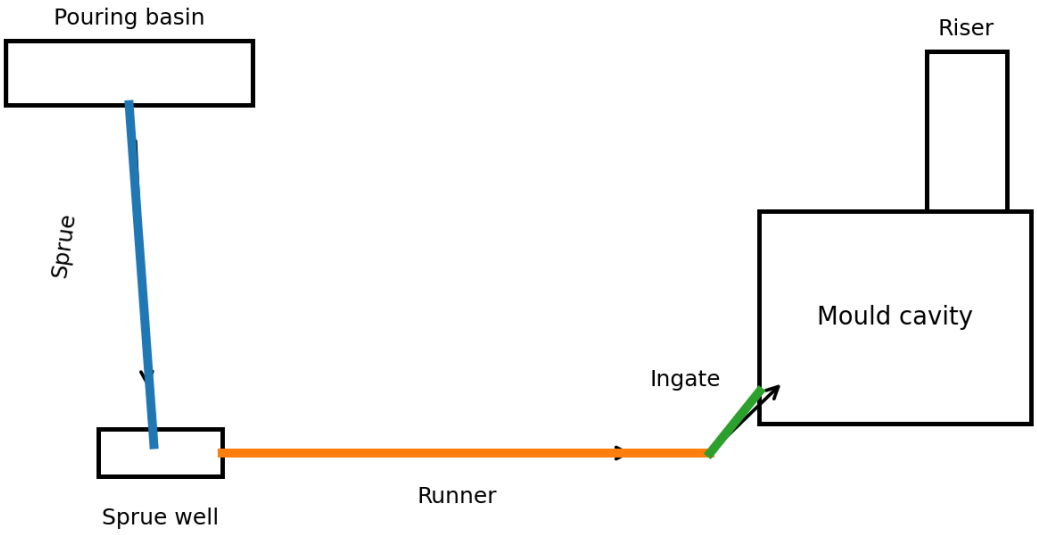
Factor	Influence
Casting mass	Heavier castings require a greater total quantity of metal and hence an appropriate higher flow capacity.
Minimum section thickness	Thin sections require quicker filling before freezing.
Alloy fluidity	Highly fluid alloys can tolerate lower flow rates than alloys that freeze rapidly.
Mould and core strength	Very high velocities may erode weak mould/core surfaces.
Gating ratio and choke area	These directly influence the discharge rate through the system.
Pouring temperature	Higher superheat increases available filling time, but excessive temperature is undesirable.

EXAM POINT: In gating design, the desired pouring time is converted into the required metal flow rate, and the flow rate is then used to size the choke and the remaining gating passages.

3. Gating System

A gating system is the complete network of passages through which molten metal flows from the pouring vessel to the mould cavity. Its purpose is not only to carry the metal, but to control its velocity, direction, turbulence, slag entry and rate of mould filling.

Basic Elements of a Gating System



3.1 Functions of a good gating system

- Fill the mould cavity completely within the required pouring time.
- Deliver metal with minimum turbulence and aspiration of air.
- Prevent slag, dross and loose sand from entering the cavity as far as possible.
- Avoid erosion of mould and cores.
- Minimize heat loss during flow.
- Promote favourable temperature distribution and directional solidification where possible.
- Provide economical yield by avoiding unnecessarily large runners and gates.

3.2 Elements of a gating system

Element	Purpose / description
Pouring basin or cup	Receives the metal from the ladle, steadies the stream and guides it into the sprue.
Strainer / filter (when used)	Retains slag or inclusions and can reduce turbulence.
Sprue	Vertical or inclined passage carrying metal downward from the pouring basin.
Sprue base well	Enlarged pocket at the bottom of the sprue that reduces the momentum of the falling stream and turns the flow into the runner.
Runner	Horizontal passage that distributes metal to one or more ingates.
Runner extension	Continuation beyond the last ingate, helping trap first metal, slag and loose sand.
Ingate (gate)	Passage connecting runner to mould cavity; controls entry location and local velocity.
Choke	Smallest effective cross-sectional area in the gating system; largely controls discharge rate.
Riser (not strictly a gate passage)	Reservoir supplying liquid metal to compensate solidification shrinkage.

3.3 Important design considerations

- The sprue is usually tapered so that the flowing stream remains full and air aspiration is reduced.
- Sharp corners should be avoided because they cause separation, turbulence and erosion.
- The ingates should introduce metal at locations that encourage smooth filling and proper feeding.
- The choke area must be consistent with the required pouring rate.
- Runner extensions, skimming arrangements and filters may be used to reduce inclusions.

4. Gate Ratio (Gating Ratio)

Gate ratio is the ratio of the total cross-sectional areas of the main gating elements, commonly written as:

Sprue area : Total runner area : Total ingate area

4.1 Pressurized gating system

In a pressurized system, the downstream total flow area is intentionally restricted so that the passages tend to run full and a back pressure is developed. It provides compact gating and high yield, but velocities and turbulence can be higher if not carefully designed.

- The total ingate area is comparatively small relative to the upstream passage areas.
- Commonly used for ferrous castings where a compact gating system is desirable.

4.2 Unpressurized gating system

In an unpressurized system, the flow area generally increases from sprue to runners and ingates. Metal velocity is reduced, giving smoother filling and less turbulence, though the gating system occupies more volume.

- The total runner and gate areas are comparatively large.
- Frequently preferred for alloys that are very sensitive to oxidation and turbulence, such as many aluminium alloys.

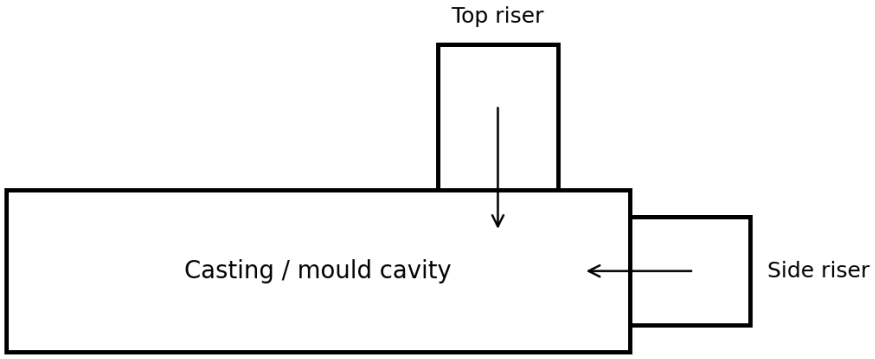
Feature	Pressurized gating	Unpressurized gating
Area tendency	Flow area becomes restricted downstream.	Flow area increases downstream.
Back pressure	Relatively higher.	Relatively lower.
Metal velocity	Generally higher.	Generally lower.
Turbulence tendency	Higher if poorly designed.	Lower; smoother filling.
Metal yield	Usually higher because passages are smaller.	Usually lower because passages are larger.
Typical use	Many ferrous castings.	Many light-alloy castings.

EXAM POINT: A gate ratio is not a universal fixed number. The chosen ratio depends on alloy, mould, casting geometry and the required filling behaviour.

5. Risers (Feeders)

A riser is a reservoir of molten metal provided in the mould to feed the casting during solidification. As the casting metal contracts while changing from liquid to solid, the riser supplies additional liquid metal to prevent shrinkage cavities and internal shrinkage porosity.

Typical Riser Locations



Risers should feed the last-to-solidify (hot-spot) regions.

5.1 Types of risers

Basis	Type	Description
Exposure	Open riser	Top of riser is open to atmosphere; simpler but loses heat readily.
Exposure	Blind (closed) riser	Completely enclosed by moulding sand; often retains heat better.
Position	Top riser	Placed above the casting; gravity assists feeding.
Position	Side riser	Placed beside the casting and connected through a neck.
Function / pressure	Atmospheric riser	Conventional open riser fed under atmospheric pressure.
Function / pressure	Blind riser with pressure effect	Gas pressure and hydrostatic head can assist feeding when properly designed.
Heat conservation	Insulated / exothermic riser	Uses insulating or exothermic material to delay solidification and improve feeding efficiency.

5.2 Requirements of an effective riser

- The riser must contain enough liquid metal to compensate the solidification shrinkage of the region it feeds.
- It must remain molten longer than the portion of casting being fed; therefore its solidification time should be greater than that of the casting hot spot.
- There must be an open feeding path between riser and casting until feeding is complete.
- The riser should develop sufficient pressure head to force liquid metal into shrinking regions.
- Its neck should permit feeding but should not be unnecessarily large, because the riser must later be removed.
- The riser should be as small as practical while still feeding safely, to maximize casting yield.

5.3 Location of risers

- Place risers near the heaviest or last-to-solidify sections (hot spots).

- Promote directional solidification from thin/cool regions toward the riser.
- For a casting with several isolated heavy sections, more than one riser may be required.
- Avoid locations where the feeding path is blocked by an early-freezing thin section.
- Where possible, locate the riser on a surface that can be cleaned or machined easily.

EXAM POINT: Riser placement is based on solidification sequence: the casting should freeze progressively toward the riser, and the riser should freeze last.

6. Chills

Chills are high-thermal-conductivity materials placed in or against a mould to increase the local rate of heat extraction. They are used to control the solidification pattern, eliminate hot spots, improve directional solidification and reduce shrinkage defects.

6.1 Types of chills

Type	Description	Typical purpose
External chill	Metal block placed against the mould cavity surface; it does not become part of the casting.	Rapidly cools a heavy local section and shifts the last-freezing region toward a riser.
Internal chill	Metal insert placed inside the mould cavity and fused into the casting during pouring.	Provides local chilling where an external chill cannot reach; must be metallurgically compatible and clean.
Natural chill	A chill effect obtained from the casting geometry or a metallic mould/core without a separate inserted chill.	Used where the normal mould arrangement already provides locally rapid heat extraction.

6.2 Requirements and precautions

- A chill should have adequate thermal conductivity and heat capacity.
- Internal chills must be clean, dry and compatible with the casting alloy.
- Improperly placed chills may create excessive thermal gradients, hard spots or cracks.
- The chill should cooperate with the riser system to obtain directional solidification.

7. Special Casting Process - Permanent Mould Casting

Permanent mould casting uses a reusable metallic mould, commonly made of cast iron, steel or another heat-resistant metal. Unlike sand casting, the mould is not destroyed after each casting.

7.1 Basic procedure

1. The metal mould is cleaned and a refractory or lubricating coating is applied.

2. The mould is preheated to a suitable temperature.

3. Cores, if required, are positioned. Metal cores or expendable sand cores may be used depending on shape.

4. The mould halves are closed and clamped.

5. Molten metal is poured, usually by gravity in gravity permanent mould casting.

6. The metal solidifies rapidly because the metallic mould extracts heat efficiently.

7. The mould is opened and the casting is ejected; the mould is then prepared for the next cycle.

7.2 Advantages

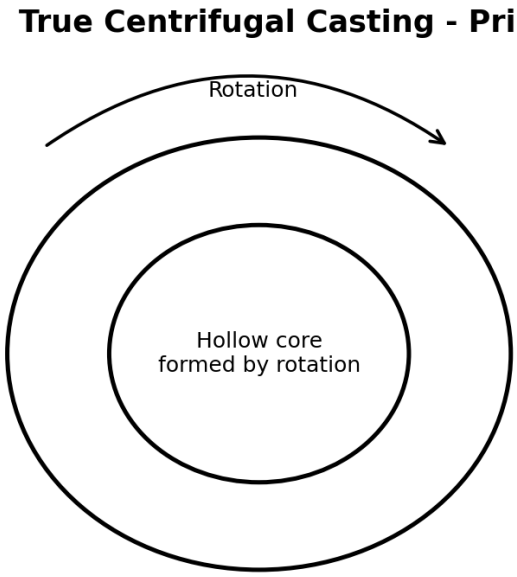
- Better surface finish and dimensional accuracy than ordinary sand casting.
- Higher production rate because the mould is reusable.
- Finer grain structure and often improved mechanical properties due to rapid cooling.
- Less fettling and cleaning.
- Good process repeatability.

7.3 Limitations and applications

- High initial mould cost, so the method is economical mainly for medium-to-large production quantities.
- Very intricate undercuts are difficult unless special cores/slides are used.
- Common for non-ferrous alloys such as aluminium, magnesium and copper-base alloys.
- Typical products include pistons, cylinder heads, cookware, small housings and automotive components.

8. Special Casting Process - Centrifugal Casting

In centrifugal casting, the mould rotates while molten metal is introduced. Centrifugal force causes the metal to move outward and press against the mould wall. Depending on the method, the process may produce hollow axisymmetric parts or a number of smaller castings arranged around an axis.



Molten metal is forced outward against the mould wall by centrifugal force.

8.1 Types of centrifugal casting

Type	Principle	Typical products
True centrifugal casting	Molten metal is poured into a rotating mould and forms a hollow cylindrical casting without a core.	Pipes, tubes, cylinder liners, rings, bushes.

Semi-centrifugal casting	The casting is generally solid and axisymmetric; centrifugal action improves feeding toward the outer regions.	Wheels, pulleys, gear blanks.
Centrifuging	Several mould cavities are arranged around a central sprue; rotation forces metal radially into the cavities.	Small irregular castings produced in groups.

8.2 Working of true centrifugal casting

1. A cylindrical mould is mounted horizontally or vertically and rotated at a controlled speed.

2. Molten metal is introduced near the axis.

3. Centrifugal force throws the liquid metal toward the mould wall.

4. Solidification proceeds generally from the outside surface inward because the mould wall extracts heat.

5. Lower-density inclusions and gases tend to concentrate nearer the inner surface, which may be machined away.

6. After solidification, rotation is stopped and the casting is removed.

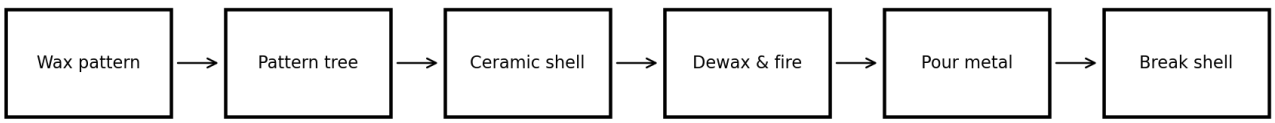
8.3 Advantages and limitations

- No core is required for many hollow cylindrical castings.
- Dense, sound structure can be obtained because of pressure generated by rotation.
- Good mechanical properties and relatively clean outer metal are possible.
- Equipment and speed control add cost; the method is mainly suited to shapes compatible with rotation.
- Inner surface may require machining to remove segregated impurities and obtain exact dimensions.

9. Special Casting Process - Investment Casting

Investment casting, also known as the lost-wax process, uses an expendable wax or similar pattern surrounded by a refractory ceramic shell. The pattern is removed by heating, leaving a precise cavity into which molten metal is poured.

Investment Casting - Process Sequence



9.1 Process steps

1. Produce accurate wax patterns in a metal die.

2. Attach several wax patterns to a central wax sprue to form a pattern tree or cluster.

3. Dip the cluster in refractory slurry and coat it with fine refractory grains (stucco).

4. Repeat dipping and drying until a shell of adequate thickness is formed.

5. Heat the shell to melt/remove the wax (dewaxing).

6. Fire or preheat the ceramic shell to develop strength and remove residual wax/moisture.

7. Pour molten metal into the hot shell.

8. Allow the metal to solidify, break away the ceramic shell, cut castings from the tree and finish them.

9.2 Advantages

- Excellent dimensional accuracy and surface finish.
- Very intricate shapes, thin sections and fine details can be produced.
- No parting line is required in the ceramic shell.
- Suitable for alloys that are difficult to machine.
- Near-net-shape production reduces machining and material waste.

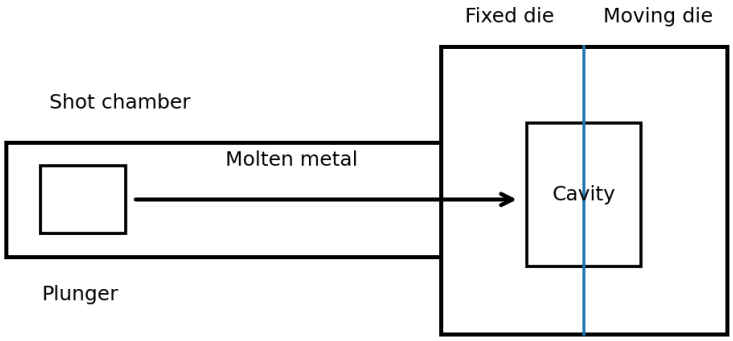
9.3 Limitations and applications

- Process is relatively expensive and time-consuming because patterns and shells are expendable.
- Casting size is commonly limited compared with ordinary sand casting, though modern processes can produce larger parts.
- Used for turbine blades, precision aerospace parts, surgical/medical components, jewellery, small tools and complex engineering components.

10. Special Casting Process - Die Casting

Die casting is a permanent-mould process in which molten metal is forced into a metallic die cavity under pressure. It is characterized by rapid production, good surface finish, close dimensional accuracy and the ability to produce thin-walled intricate parts.

Pressure Die Casting - Basic Arrangement



10.1 Main parts of a die-casting arrangement

- Fixed die half and moving/ejector die half.
- Die cavity and cores/slides where needed.
- Metal injection system or shot chamber.
- Plunger/piston for forcing metal into the cavity.
- Ejector pins for removing the solid casting.
- Cooling passages and vents/overflows.

10.2 Hot-chamber die casting

In the hot-chamber process, the injection mechanism is immersed in or directly connected to a bath of molten metal. A plunger forces a measured charge through a gooseneck/nozzle into the die. The cycle is very fast.

- Suitable mainly for low-melting alloys that do not attack the submerged injection system severely, such as zinc and some magnesium/tin/lead alloys.
- Advantages: very high cycle rate and less metal handling.

10.3 Cold-chamber die casting

In the cold-chamber process, molten metal is ladled into a separate shot chamber for each cycle. A plunger then forces the metal into the die at high pressure.

- Used for alloys with higher melting temperature or greater tendency to attack the injection equipment, notably aluminium and many copper-base alloys.
- Cycle time is slower than hot-chamber because the metal must be transferred into the shot sleeve.

Feature	Hot-chamber	Cold-chamber
Metal supply	Injection system connected to/immersed in molten metal.	Metal ladled into shot chamber each cycle.
Cycle speed	Faster.	Slower.
Typical alloys	Zinc and other suitable low-melting alloys.	Aluminium and many higher-temperature alloys.
Equipment exposure	Injection system directly exposed to melt.	Injection system has less continuous immersion.

Main advantage	Very high productivity.	Handles alloys unsuitable for hot chamber.

10.4 Advantages of die casting

- Very high production rate.
- Excellent surface finish and dimensional repeatability.
- Thin walls and intricate shapes are possible.
- Little machining may be required.
- Inserts can sometimes be cast in place.

10.5 Limitations

- Dies and machines are expensive, so large production quantity is normally necessary.
- Casting size is limited by machine locking force and shot capacity.
- Entrapped air and gas porosity can occur if filling and venting are not properly controlled.
- The process is mainly used for non-ferrous alloys.

11. Comparison of Special Casting Processes

Process	Mould	Pressure / force	Accuracy & finish	Production rate	Typical strength
Permanent mould	Reusable metal mould	Mainly gravity	Good	Medium-high	Good due to rapid cooling
Centrifugal	Rotating metal/refractory mould	Centrifugal force	Good for suitable shapes	Medium-high	Dense structure, especially outer region
Investment	Expendable ceramic shell	Gravity / assisted variants	Excellent	Low-medium	Excellent detail; alloy dependent
Die casting	Reusable steel die	High injection pressure	Excellent	Very high	Good; porosity control is important

Topic	One-line revision point
Optimum pouring temperature	Enough superheat for complete filling without creating excessive oxidation, gas pickup or shrinkage problems.
Pouring time	Time required to fill the cavity; too slow causes freezing, too fast causes turbulence/erosion.
Gating system	Network that guides and controls molten metal from ladle to mould cavity.
Gate ratio	Sprue area : runner area : ingate area.
Pressurized gating	Restricted downstream area; passages tend to run full and velocity is relatively high.
Unpressurized gating	Increasing downstream area; smoother and lower-velocity flow.
Riser	Reservoir that feeds liquid metal to compensate solidification shrinkage.

Chill	High-conductivity body used to accelerate local solidification.
Permanent mould casting	Reusable metallic mould, usually gravity filled.
True centrifugal casting	Rotating mould creates hollow axisymmetric part without a core.
Investment casting	Lost-wax process using a refractory ceramic shell.
Die casting	Molten metal is injected into a reusable metal die under pressure.

- Define pouring temperature and explain why an optimum value is required.
- What is pouring rate? Explain the defects caused by excessively slow and excessively fast pouring.
- Draw a gating system and explain each element.
- Define gate ratio. Distinguish between pressurized and unpressurized gating systems.
- What is a riser? State the requirements and rules for riser location.
- Differentiate between open and blind risers.
- Define chills. Differentiate between external and internal chills.
- Explain permanent mould casting with advantages and limitations.
- Explain true centrifugal, semi-centrifugal and centrifuging processes.
- Explain investment casting step-by-step.
- Differentiate hot-chamber and cold-chamber die casting.
- Compare permanent mould, centrifugal, investment and die casting processes.

Reference

Primary reference framework: S. K. Hajra Choudhury and A. K. Hajra Choudhury, Elements of Workshop Technology, Vol. I: Manufacturing Processes, sections dealing with foundry practice and special casting processes. These notes are paraphrased and reorganized for classroom study and examination revision.